



Instructor: ..... Date: .....

Learner: ..... Regional Branch: .....

Program: ..... Center: .....

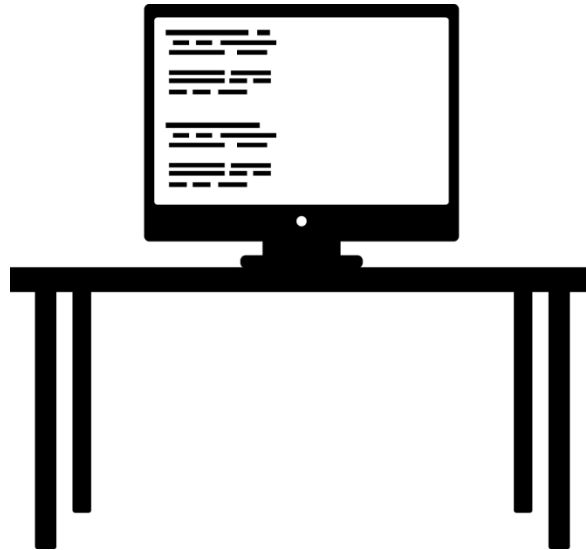
## CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



### Let's Start!

#### Understanding the Char Data Type



#### Learning Outcome:

Talk about principles, concepts, characteristics of general-purpose computer programming language in spoken and written situations.

#### Ask your partner:

- What is the strangest character you know in a programming language?
- How different is the Java application in other countries? Explain.

#### Can Do Statements:

- I can say what char data type means in Java.
- I can give examples on how to declare char data types.
- I can talk about the relationship between char and integer.





## Let's Explore!

Mark's giving a hand\*...

\*To "give a hand" means to help or assist someone with something.



*Instructions: Read the text below.*

**Gabriela:** Will we have our tutoring session tomorrow, Mark?

**Mark:** Sure, I can be there for you at 5. By the way, does Sally come too?

**Gabriela:** Yes. Now, are we discussing the concept of the char data type in Java tomorrow?

**Mark:** Well, that is definitely the least difficult topic to address. The most basic thing to know about that data type is that its name refers to 'character' and that it takes 16 bits of memory. Now, it basically allows you to take any character of any language or system and unify it by assigning it a numeric value, a single standard, through the Unicode encoding methodology.

**Gabriela:** I see. Now, are we talking over any important aspect of this data type?

**Mark:** Sure. You may ask yourself: now that I know about it, how am I going to declare char variables? And well, that is the most imperative question to ask since two are the answers for it. For example, when you omit single quotes, the Unicode representation will be set, whereas including the single quotes will assign a given value to the used character. Let me be more specific about this, if you code "char ninety\_three = 65;" it means that the resulting char value, which is the letter A (that corresponds to 65), will be displayed. On the other hand, if you code "char ninety\_three = '65';" the outcome will be 'A'.

**Gabriela:** Great. Are we covering the relationship between char and integer, then?

**Mark:** Will you really ask that? The relationship between these two is the simplest one as it is possible to use chars just as if they were integer values. Just as you can see, they are closely related.

**Gabriela:** I see. You are definitely the best.

**Let's Practice!**

Help Gabriela to complete the char data information in a correct way.

*Instructions: Read the paragraph. Underline all the mistakes in the paragraph.*

The char data type is the least most difficult topic to address. The more basic thing to know about char data type is that its name refers to 'character' and that it takes 16 bits of memory. It is the more easiest way to take any character of any language or system and unify it by assigning it a numeric value, a single standard, through the Unicode encoding methodology. Asking about how to declare char variables is the imperatives question in this domain.

---

---

---

---

**Let's Practice!**

Identifying some words related to char data.

*Instructions: Match column A with column B.*

**COLUMN A**

1. Bit
2. Character
3. Single quote
4. Unicode
5. Java

**COLUMN B**

- a. a programming language developed to be used in Internet environment
- b. letter, number, space, punctuation mark, or symbol typed
- c. a symbol that is used to provide characters with special values
- d. the smallest computer measurement data unit
- e. universal character encoding standard



## Extension Activity

Let's build some examples to get the main char data features.

*Instructions: Complete the following steps. Fill in the Self-Assessment.*

- Look for information about char data implementation in Java examples.
- Write notes and design a poster in which you explain the different in functionality from other types of data.
- Share with your teacher and ask all his questions.

### Self-Assessment

Yes

No

- I look for information about char data examples in Java.
- I write notes about data chat type.
- I design poster to explain different aspects of char data.
- I talk about char data type in java.

☐ ☐
☐ ☐
☐ ☐
☐ ☐


## Target Words

- **tutoring session** (noun): a private academic session that aims to clarify doubts regarding the topics of a class  
e.g. Will we have our tutoring session tomorrow, Mark?
- **refer** (verb): to allude or indicate something  
e.g. The most basic thing to know about that data type is that its name refers to 'character' and that it takes 16 bits of memory.
- **character** (noun): a letter or number that is part of a given language or system  
e.g. Now, it basically allows you to take any character of any language or system and unify it by assigning it a numeric value, a single standard, through the Unicode encoding methodology.
- **imperative** (adjective): of extreme importance  
e.g. And well, that is the most imperative question to ask since two are the answers for it.
- **single quote** (noun): symbol that is present in the coding keyboard ( ' ' ) and that is used to provide characters with special values  
e.g. For example, when you omit single quotes, the Unicode representation will be set, whereas including the single quotes will assign a given value to the used character.
- **cover** (verb): to include something within the topics to be revised in a class  
e.g. Are we covering the relationship between char and integer, then?



## Learning Points

### The Future with the Present Continuous in Questions

You use the **Present Continuous** in its interrogative form to ask special questions.

You can use the Present Continuous tense to express scheduled plans in the near future. Regarding the interrogative form of this tense, the structure that we follow depends on the type of question you want to ask.

For open questions:

Wh + am/is/are + subject + verb (-ing form) + complement + ?

**What are you studying** tomorrow night, char data types or comments in Java?

For yes /no questions:

Am/Is/Are + subject + verb (- ing form) + complement

**Are we talking** over this data type tomorrow?

Note that both examples have a time expression that helps us to understand the future context and differentiate the present from the future.

#### Example

- *Now, are we talking over any important aspect of this data type?*
- *Are we discussing the concept of the char data type in Java tomorrow?*

### Understanding the Unicode Computer Encoding Methodology

The Unicode Computer Encoding Methodology unifies all the different languages and special characters into a single standard.

You use the Unicode encoding methodology in order to assign a sole number for each and every character of the system. In other words, numeric values are given to each character no matter the language, the computer system, or the computer platform. Furthermore, all these characters and their numeric values are unified by their numeric representation. This numeric representation will vary depending on the type of character; for instance, the same letter 'G' and 'g' (with different capitalization) would get different numeric representations.

For example,

*The letter 'A' is represented with a numeric value of 65, while the letter 'J' is with 74.*

#### Example

- *The letter 'A' is represented with a numeric value of 65, while the letter 'J' is with 74.*

